Q1. Load the CSV file into a Pandas DataFrame. Report the number of rows and columns, inspect the data types, and identify which columns are numerical, categorical, date-related, and identifier fields.

Q2. Convert **Order_Date** into a proper datetime column. Find the earliest and latest order dates, and calculate the total number of days covered by the dataset.

Q3. Use NumPy and Pandas to calculate the mean, median, standard deviation, minimum, and maximum for **Net_Sales_INR**, **Profit_INR**, **Quantity**, and **Delivery_Days**. Compare the mean and median and comment on any noticeable differences.

Q4. Create a new column named **Profit_Margin_Percent** using profit and net sales. Identify orders with negative, zero, and positive profit margins, if any.

Q5. Create the following date-based columns: **Order_Year**, **Order_Month**, **Month_Name**, and **Quarter**. Use them to prepare the DataFrame for time-based analysis.

Q6. Create a **Discount_Category** column by defining sensible discount bands of your choice. State the logic used and calculate the number and percentage of orders in each band.

Q7. Create an **Order_Value_Category** column using your own meaningful thresholds for low-, medium-, and high-value orders. Explain the thresholds and count the orders in each category.

Q8. Calculate total gross sales, total discount amount, total net sales, and total profit. Also calculate the overall discount rate as total discount divided by total gross sales.

Q9. Which product category generates the highest net sales? Show net sales by category in descending order and visualize the comparison with a Matplotlib bar chart.

Q10. Which five products generate the highest total net sales? Display the results in a sorted table and create a horizontal bar chart using Seaborn.

Q11. Compare total net sales across the five regions. Identify the strongest and weakest region by net sales and visualize the results.

Q12. Which cities contribute the most to net sales? Find the top 10 cities and visualize their contribution using a horizontal bar chart. Add clear axis labels and a descriptive title.

Q13. Compare the average order value across customer segments. Which segment has the highest average order value, and how does its order count compare with the other segments?

Q14. Calculate each sales channel's share of total net sales. Visualize the channel contribution using a suitable chart and explain which channel contributes the most.

Q15. Identify the most frequently purchased product category by order count. Compare its order count with its contribution to net sales. Does the most popular category also generate the most revenue?

Q16. Calculate total profit and average profit for every product category. Identify categories with high sales but comparatively low average profit.

Q17. Find the 10 most profitable products based on total profit. Visualize the ranking and compare it with the top products by net sales.

Q18. Analyze the relationship between **Discount_Percent** and **Profit_INR**. Create a Seaborn scatter plot and describe the pattern you observe.

Q19. Compare average profit margin across discount categories. Does a higher discount band correspond to a lower average profit margin? Support your observation with a visualization.

Q20. Which payment method is associated with the highest total net sales? Also calculate its order count and average order value. Present the comparison in a chart.

Q21. Compare customer ratings across product categories. Use a Seaborn boxplot to show the distribution of ratings and identify categories with relatively high or low ratings.

Q22. Investigate whether order quantity is associated with net sales. Create a scatter plot, calculate the correlation coefficient, and explain what the result indicates.

Q23. Calculate the percentage of orders marked as **On Time** and **Delayed**. Visualize the delivery-status distribution using a count plot or bar chart.

Q24. Compare average customer ratings for on-time and delayed orders. Quantify the difference and visualize it using a suitable Seaborn chart.

Q25. Which shipping mode has the shortest average delivery time? Compare average delivery days and delayed-order percentage across shipping modes.

Q26. Examine the relationship between delivery days and customer ratings. Create a boxplot or point plot and describe whether ratings tend to change with delivery time.

Q27. Find the regions with the highest delayed-order percentage. Calculate the delay rate as delayed orders divided by total orders in each region, then visualize the regional comparison.

Q28. Calculate monthly net sales and monthly profit. Create a two-panel Matplotlib visualization showing how both measures change over time.

Q29. Identify the best and worst month by total net sales across the full dataset. Report the month and year, the sales value, and the difference between the two months.

Q30. Compare quarterly net sales across years. Use a grouped bar chart or line chart and comment on any visible seasonal or year-over-year pattern.

Q31. Create a Seaborn heatmap of monthly net sales, with years as rows and months as columns. Which month-year combination has the highest sales, and what patterns are visible?

Q32. Build a final business dashboard-style figure containing at least four different visualizations: (a) monthly net sales trend, (b) net sales by category, (c) profit by region, and (d) delivery-status distribution. Use appropriate titles, labels, legends, and a consistent visual style. Add a short written conclusion containing at least five evidence-based business insights.